

Linux On Old Hardware

The full written lesson — distro & install, command line fundamentals, systemd services, resource pressure & recovery.

1. Choosing a distro deliberately, and installing under constraints

Old hardware is the best Linux teacher because you can't be careless with resources.

- Lightweight distros: Debian's minimal net-install, Xubuntu (XFCE), Alpine (musl+busybox, real compatibility quirks for desktop use).
- The desktop environment matters more than the distro name — GNOME/KDE can idle over a gigabyte before you've opened anything.

The trap: installing a heavy desktop environment on limited RAM. XFCE, LXQt, or a bare window manager leaves real headroom instead of consuming most of a 2GB machine at idle.

Worked: the same physical machine that "felt too slow" often runs perfectly well once OS overhead itself is what changes — the hardware was rarely the real bottleneck.

2. Command line fundamentals: permissions and processes

Two habits carry most of the real risk in daily Linux use.

- Permissions (rwx) as octal (755) or symbolic (rwxr-xr-x). `chmod` changes permissions, `chown` changes ownership.
- `kill` sends SIGTERM (catchable, allows cleanup); `kill -9` sends SIGKILL (immediate, no cleanup chance).

The trap: running everything as root. A regular user plus sudo for specific commands limits the blast radius of an ordinary mistake.

Worked: a process killed with -9 mid-write gets zero chance to flush buffers — plain kill lets it shut down properly first.

3. Services with systemd: auditing what starts at boot

Every enabled service on constrained hardware is wasted RAM/CPU/boot time if it isn't needed.

- `systemctl list-unit-files --state=enabled` audits everything set to auto-start.
- `systemctl disable --now <service>` disables at boot AND stops it right now, one command.

The trap: enabling services you don't need and never checking. Packages enable services as install side effects; nobody audits the list later.

Worked: a fresh install often has Bluetooth and printing enabled with no Bluetooth hardware or printer ever attached — pure removable overhead.

4. Diagnosing resource pressure, and recovering a machine that won't boot

Turning "feels slow" or "won't boot" into a diagnosable problem instead of a mystery.

- `top` / `htop` — CPU/RAM per process. `df -h` — disk space. `journalctl` — why something failed.
- Heavy swap usage is often the real "feels slow" cause — disk-backed overflow memory is dramatically slower than RAM.
- Boot recovery: rescue media → mount the broken root partition → `chroot` in → fix config → exit → reboot.

The trap: no recovery plan before experimenting. Have rescue media ready and know the procedure before deliberately breaking something.

Worked: a broken `/etc/fstab` entry can prevent boot entirely — fixed from rescue media in minutes, no reinstall needed.